

# SHORT VS LONG INTERVALS IN TRAINED CYCLISTS

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High intensity Training (HIT) is a widely used strategy to stimulate positive physiological adaptations for endurance performance. It represents performing intervals at an intensity above the second threshold/critical power/FTP.

To get the best results from your HIT, the general recommendation is spending as much time as possible at an oxygen consumption rate  $>90\%$  of the  $VO_{2max}$  (maximum oxygen consumption).

While many different HIT typologies are being used in the field, there are few studies which tried to understand which is the best to improve performance.

***Which high intensity training format is the best to improve performance?***

***Short Intervals or Long Intervals?***

A study published in 2015 by Prof. Bent Rønnestad and Colleagues (Lillehammer University College, Norway) on *Scandinavian Journal of Medicine & Science in Sports* tried to answer this question.

## HOW DID THEY TRAIN?

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- 20 trained but not professional cyclists (~33 years old and ~66 VO<sub>2</sub>max) were divided into two groups: SI (Short Intervals) and LI (Long Intervals).

- For the following 10 weeks the two groups performed the same training program:

8-10 hrs of low intensity training + 2x week HIT sessions.

The only difference between the two groups was about the HIT format they performed:

- **SI group performed Short intervals HIT sessions:**

**3x13' alternating 30" on/15" off with 3' recovery in between.**

- **LI group a Long Intervals HIT sessions:**

**4x5' on/2.5' off.**

15-20min of warm-up and cool down were included in each session.

If you make some calculations, you can see as both SI and LI groups accumulated a similar time in the 'on' phase: 19min30s vs 20 min (3x13x30" vs 4x5').

**In both HIT's formats, the 'on' phases were performed at the maximum sustainable effort to obtain the highest average power throughout all the intervals.**

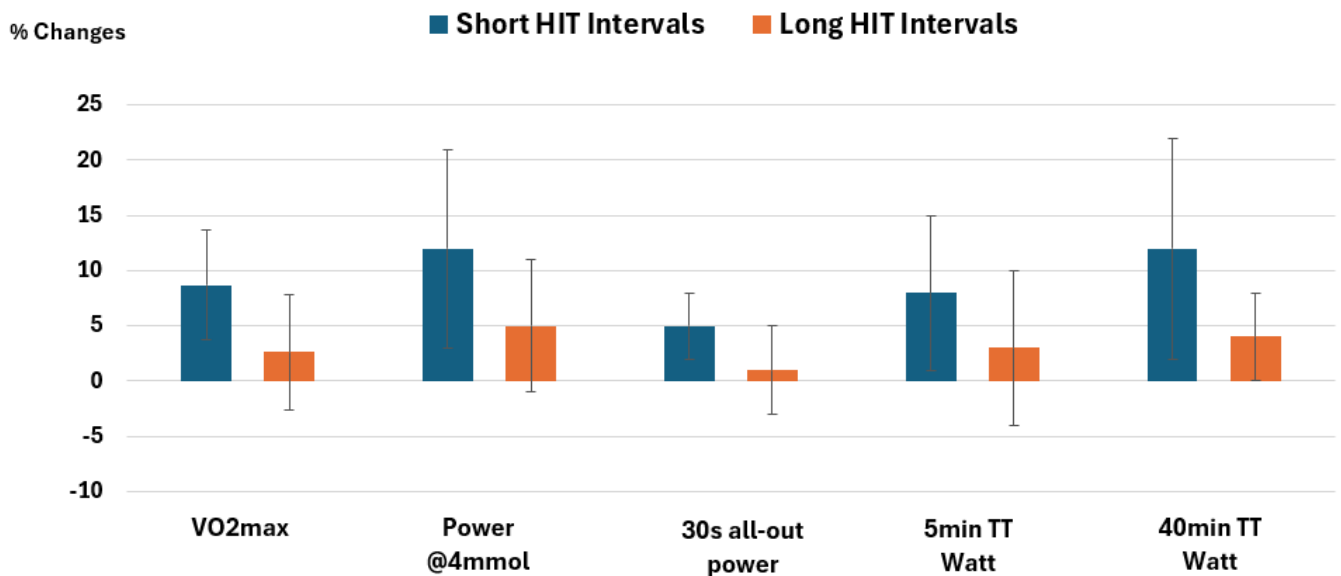
The 'off' phases were performed at 50% of the power output sustained during intervals, so it was active recovery (read high zone 1 or low zone 2).

## WHAT DID THEY FOUND?

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**Globally, after the 10 training weeks. the SI group improved more all the physiological and performance parameters compared to the LI group:**

- VO<sub>2</sub>max: 8.7±5.0% vs. 2.6±5.2%
- Wmax (power output in the last step of a maximal incremental test): 8.5±5.2% vs 1.6±3.6%
- Power output at 4mMol/L of blood lactate concentration: 12±9% vs 5±6%
- 40-min time trial: 12±10% vs 4±4%
- 5-min time trial: 8±7% vs 3±7%
- 30s all-out-trial (Wingate Test): 5±3% vs 1±4%



% changes between pre and post the 10 training weeks in physio and performance parameter in the Short vs Long HIT Intervals groups. TT is time trial.

Note that, as a result of performing the intervals at the maximum sustainable power output, **the SI group average power during the 'on' phases was higher compared to the LI group: 363 W (~120% second threshold) vs 324 W (~108% second threshold).** Despite this, a similar rating of perceived exertion (RPE) between groups, both when averaged across all the intervals and in the last interval, was reported.

## PRACTICAL TAKEAWAYS

In trained but not professional cyclists, 3x13min alternating 30s on and 15s off seems to provide superior physiological and performance adaptations compared to 4x5min intervals when both formats are performed at the maximum sustainable intensity.

## GABRIELE'S COMMENT

- These data suggest us that short HIT intervals seems to be a better solution to improve performance when compared to long intervals for duration ranging from 30s to 40min.

What is the underlying physiological mechanism? We don't have a definite answer, but the observed differences could be related to the fact that, **when matching the perceived exertion, short intervals permit to accumulate more time at higher power output (and so maybe to stimulate a stronger biological signal for physiological adaptations) when compared to the long intervals.** However, this remains just a speculation with no data to support it.

- A few points when applying these data on the field:
  - **Individualize**: as every athlete is unique, I would suggest trying both HIT formats for each individual and then looking at which one he/she can sustain a higher power output and heart rate for a given total duration and perceived effort. This is probably the HIT most effective for him/her. Anyway, it is very likely that it will be the short intervals' case.
  - **Progressive overload**: when starting to insert short intervals in the training plan, it could be reasonable to start with a lower number of repetition than the 3x13min used in this study (e.g. 3x8'), and then progressively increase the number of repetitions workout after workout to arrive smoothly at the 3x13' used in this study. You could also start with a 1:1 work:recovery ratio instead of 2:1 (so alternating 30s 'on' and 30s 'off', instead of 30s 'on' and 15s 'off').
  - **Specificity**: even if short intervals seem generally better to obtain adaptations both for short and long duration performance, it could make sense to use long intervals when they are very similar to the effort that will be performed in competition. For example the track individual pursuit's effort is very similar to long intervals.

This study was performed on trained but non professional athletes.

Can the same results be applied on pro cyclists? We will see this in the next posts!

Thanks for Reading! Spread with your Team!

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## REFERENCES:

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1. Rønnestad BR, Hansen J, Vegge G, Tønnessen E, Slettaløkken G. Short intervals induce superior training adaptations compared with long intervals in cyclists - an effort-matched approach. Scand J Med Sci Sports. 2015 Apr;25(2):143-51.



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